

JavaScript Basics Assignment

Instructions

- Answer all questions.
 - Write code where required.
 - Use proper indentation and comments in your code.
 - Total Questions: 30
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Part A — Theory Questions

1.

Explain the difference between `==` and `===` in JavaScript with examples.

2.

What are JavaScript primitive data types? Explain each briefly.

3.

What is an array in JavaScript?

Create an array of 5 fruits and access:

- the first element
 - the last element
-

4.

What is an object in JavaScript?

Create an object for a student containing:

- name
 - age
 - grade
-

5.

Write the syntax of an `if...else` statement.

Create a program that checks whether a number is positive or negative.

6.

What is a loop?

Name and explain:

- `for` loop
 - `while` loop
-

7.

Explain the difference between a `for` loop and a `while` loop.

8.

What is the purpose of `console.log()`?

9.

How do you select an HTML element using JavaScript?

Write examples using:

- `getElementById()`
 - `querySelector()`
-

10.

What is an event in JavaScript?
Give 3 examples of events.

11.

Explain the difference between `null` and `undefined`.

12.

What is the use of the `typeof` operator?
Write examples using different data types.

13.

What is a callback function?
Write a simple example.

Part B — Arrays & Strings

14.

Explain the difference between `push()` and `pop()` with examples.

15.

What is the use of `shift()` and `unshift()` methods?

16.

Write a program to find the length of a string entered by the user.

17.

What is string concatenation?

Write a program that joins first name and last name.

18.

Explain the difference between `slice()` and `splice()` with examples.

19.

Write 3 different ways to convert a string into a number in JavaScript.

20.

What is `NaN` in JavaScript?

Give one example that produces `NaN`.

Part C — Functions & Scope

21.

What is the purpose of the `return` statement in a function?

22.

What is scope in JavaScript?

23.

Explain the difference between:

- global scope
- local scope

Provide examples.

24.

What is hoisting in JavaScript?

25.

What are default parameters in JavaScript functions?

Write an example.

Part D — Modern JavaScript

26.

What is the spread operator (`...`) in JavaScript?

Write examples with:

- arrays
 - objects
-

27.

Explain the difference between `includes()` and `indexOf()`.

28.

What is JSON? Why is it used?

29.

How do you convert a JavaScript object into JSON?

Write an example using `JSON.stringify()`.

Part E — DOM & Events

30.

Explain the difference between:

- `parseInt()`
 - `Number()`
-

31.

What is the use of `addEventListener()`?

Write a button click example.

32.

Explain the difference between:

- `innerHTML`
 - `textContent`
-